Homework 10

The `chardev2.c` code is a Linux kernel module that creates a character device (`/dev/char_dev`)..

Key Components

This kernel module adds a driver device "`/dev/char_dev`" which the user can write to and read from. The module additionally logs how often the user has read from this device file and outputs this log to the `dmesg` log.

Device Operations

The module defines several device operations:

- `device_open`: Called when a process attempts to open the device file. It increments a module reference count.
- `device_release`: Called when a process closes the device file. It decrements the module reference count.
- `device_read`: Reads data from the device into user space.
- `device_write`: Writes data from user space into the device.
- `device_ioctl`: Handles various ioctls, including setting and getting messages and retrieving the nth byte.

Module Initialization and Cleanup

- `chardev2_init`: Initializes the module, registers the character device, and creates the corresponding device file in `/dev`.
- `chardev2_exit`: Cleans up resources, unregisters the character device, and destroys the device file.

Device Registration and User Interaction

The module registers a character device with the kernel using `register_chrdev`. Additionally, it creates a device class and device file in `/dev` using the `class_create` and `device_create` functions.

User-space processes can interact with the device by performing standard file operations on `/dev/char_dev`. Reading and writing to the device trigger corresponding functions in the module, allowing data exchange between the kernel and user space.

Console Output

Output of running `sudo dmesg` which shows all the logged output of the kernel module including the number of times the file has been read from.

We wrote to the file a number of times and then read from the file 3 times, which can be seen in the highlighted sections of the following terminal output.

```
[ 4943.073299] Device created on /dev/char_dev
[ 4963.312286] device_open(0000000053d9ea09)
[ 4963.312315] device_release(000000005b35325e,0000000053d9ea09)
[ 4998.326935] device_open(00000000ed093702)
[ 4998.326946] device_write(00000000ed093702,00000000504a739b,3)
[ 4998.326950] device_release(000000005b35325e,00000000ed093702)
[ 5005.072055] device_open(000000005490851c)
[ 5005.072085] Read 3 bytes, 131069 left
[ 5005.072088] Char dev file read 1 times by user
[ 5005.072108] device_release(000000005b35325e,000000005490851c)
[ 5012.137849] device_open(00000000140126cf)
[ 5012.137880] Read 3 bytes, 131069 left
[ 5012.137883] Char dev file read 2 times by user
[ 5012.137903] device_release(000000005b35325e,00000000140126cf)
[ 5019.458220] device_open(0000000022e8146a)
[ 5019.458231] device_write(0000000022e8146a,00000000504a739b,4)
[ 5019.458235] device_release(000000005b35325e,0000000022e8146a)
[ 5022.073151] device_open(00000000c679f3e0)
[ 5022.073170] Read 4 bytes, 131068 left
[ 5022.073172] Char dev file read 3 times by user
[ 5022.073190] device_release(000000005b35325e,00000000c679f3e0)
```